

Modeling Crime-Solving with LLMs via Structured Knowledge Representations and Fine-Tuning

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ABSTRACT

This study investigates how organizing evidence into structured knowledge representations affects a large language model's (LLM) capacity to reason through and solve complex, multi-layered crime scenarios. An LLM was used to extract event-based entities and timestamps from raw test case files and map them into two contrasting graph structures: a knowledge graph capturing ontological connections between people, places, evidence, and facts, and a temporal graph encoding events and their time-based relationships. Both graphs were stored as JSON and stored in Neo4j for visualization and retrieval. LLM agents were then evaluated across four experimental conditions – baseline (sans graph), knowledge graph supplement, temporal graph supplement, and combined (both graphs' supplement) – upon a dataset of 24 multidisciplinary crime cases. Results indicated that the temporal graph (81.3% accuracy) outperformed both the raw LLM baseline (79.2%) and the standalone knowledge graph (75.0%). Additionally, this graph tended to have the highest self-assessed confidence level (average 90.8%). The combined graph matched the performance of its weakest component (75.0%). Subsequently, a gemini-2.5-flash model fine-tuned on step-by-step reasoning chains and trained on 80 cases (70/30 split) achieved an accuracy of 89.6% on the 24 complex test cases with an average confidence of 91.0%. These findings suggest that the structure in which information is presented to an LLM matters as much as the information itself, and that fine-tuning can substantially improve the way LLMs utilize these knowledge structures for problem-solving.

1. INTRODUCTION

Large language models have demonstrated a remarkable capacity for understanding and generating text, yet their ability to perform deep, multi-step deductive reasoning remains a challenge. Criminal investigation is an instance of a domain in which deeper reasoning is required – integrating heterogeneous evidence, pinpointing temporal contradictions, and making broader inferences across evidence of motive and opportunity.

Prior approaches to enhancing LLM reasoning in complex domains have explored retrieval-augmented generation (RAG), graph-based reasoning, and prompt engineering.

However, few studies have systematically compared the effect of different knowledge structures – specifically, ontological knowledge graphs versus time-based graphs – on LLM deductive performance, and neither have they attempted to combine both approaches with fine-tuning.

This project aimed to explore the hypothesis: *how information is structured for an LLM may be just as important as what and how much information is provided*. An LLM’s reasoning on 24 textual crime case files was tested when alone as well as when supplemented with knowledge and temporal graphs.

2. BACKGROUND & RELATED WORK

Knowledge graphs are well-established in the domain of artificial intelligence as a means of encoding semantic relationships between entities. For example, Microsoft’s GraphRAG demonstrated that supplementing RAG pipelines with graph-based knowledge organization enables it to create more coherent synthesis across various domains (Edge et al., 2024). However, standard knowledge graphs tend to emphasize static, event-based relationships like what entities exist and how they are organized while lacking the dynamic and casual structure of events through time.

Temporal reasoning has received increasing attention as a distinct challenge for LLMs. Research on temporal graphs suggests that encoding the sequence of occurrence of events can improve an AI system’s ability to reason about cause-and-effect relationships and identify contradictions in a timeline (Liang et al., 2023). This proves to be especially crucial in crime applications in order to recognize false alibis.

Fine-tuning language models on domain-specific datasets has consistently improved performance across a range of specialized tasks, from medical to legal reasoning. Combining structured inputs with targeted fine-tuning – sometimes called GraphRAFT – is a promising strategy for tasks requiring both broad knowledge retrieval and deep reasoning (Kang et al., 2025). This project builds on this notion to construct and evaluate an optimized pipeline for LLM-driven crime analysis.

3. METHODS

3.1 Dataset

A dataset of 24 multidisciplinary crime cases was constructed, each accompanied by a solution key specifying the perpetrator and the reasoning chain supporting that conclusion. Test cases were either generated using large language models (Gemini / ChatGPT) or synthesized by neutering existing online short mystery stories to remove their solutions. Each case required critical, cross-disciplinary reasoning – encompassing forensic science, chemistry, behavioral psychology, and temporal logic – to solve correctly. They were meant to simulate real-world scenarios.

The dataset was later expanded to 80 cases for fine-tuning purposes, split 70/30 into training and validation sets. An early version of a combined-graph analysis revealed a logical inconsistency in an AI-generated case (Case #3, "The SS Meridian Incident"), where two events in the solution key were temporally incompatible. This finding prompted stricter generation standards and manual review of all cases prior to evaluation.

3.2 Knowledge Graph Construction

For each case, a dedicated LLM (Gemini 2.5 Flash) was prompted to perform exhaustive entity and relationship extraction from the raw case text. Extracted data was stored in Neo4j as a labeled property graph. Node types included Person, Location, Evidence, Timestamp, Event, Alibi, and Fact. Relationship types included WAS_AT, FOUND_AT, IMPLICATES, OCCURRED_AT, PRECEDES, CONNECTED_TO, and CONTRADICTS. Each relationship was assigned a confidence score. The complete graph was then serialized to JSON and provided to the solving LLM as structured context. Cases typically yielded 110–130 nodes and 140–170 relationships.

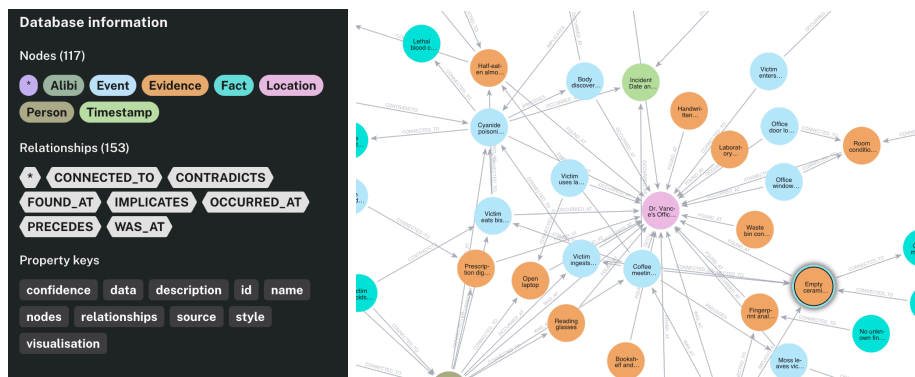
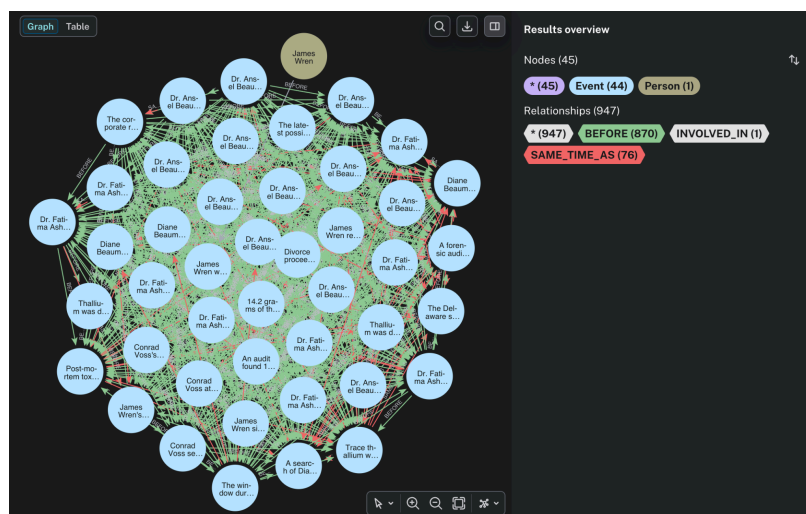


Figure 1. *Example parameters and visualization of knowledge graph*

3.3 Temporal Graph Construction

A separate LLM extraction pipeline was developed to construct a temporal graph focused exclusively on the sequence of events within a case. Node types were limited to Event and Person. Relationships encoded three temporal primitives: BEFORE, SAME_TIME_AS, and INVOLVED_IN. The LLM extracted all events involving actions, interactions, movements, discoveries, deaths, and evidence handling. Following this, it derived temporal ordering only from explicitly stated or logically inferable time references. Hallucinated edge relationships were a notable early challenge; stricter validation constraints were imposed to ensure that temporal relationships were grounded in the case text rather than inferred from background knowledge. The resulting graphs were stored as JSON and visualized in Neo4j.

Figure 2. *Sample parameters and visualization of temporal graph*

3.4 Experimental Conditions

Each of the 24 cases was evaluated under four conditions: (1) a raw baseline LLM receiving only the case text; (2) an LLM augmented with the temporal graph in JSON format; (3) an LLM augmented with the knowledge graph in JSON format; and (4) an LLM receiving both graphs simultaneously. In conditions (2) through (4), the solution prompt instructed the LLM to use the provided graph(s) to inform its reasoning. Agents were instructed to reason critically and return a top suspect along with a self-assessed confidence score from 0 to 1. All experimental conditions used Gemini 2.5 Flash.

3.5 Fine-Tuning

A fine-tuned version of Gemini 2.5 Flash was trained on the expanded 80-case dataset. For each training case, the target output was a detailed, step-by-step reasoning chain specifying the mechanism of the crime, the motive, the means, the opportunity, timeline consistency, key evidence, contradictions ruled out, and the final culprit. JSON representations of both the knowledge graph and temporal graph JSON representations were included as input context during training. The model was tuned for 4 epochs with a learning rate multiplier of 1. The fine-tuned model was subsequently evaluated on the original 24 complex test cases, producing performance metrics that were then to be compared with the original 4 models.

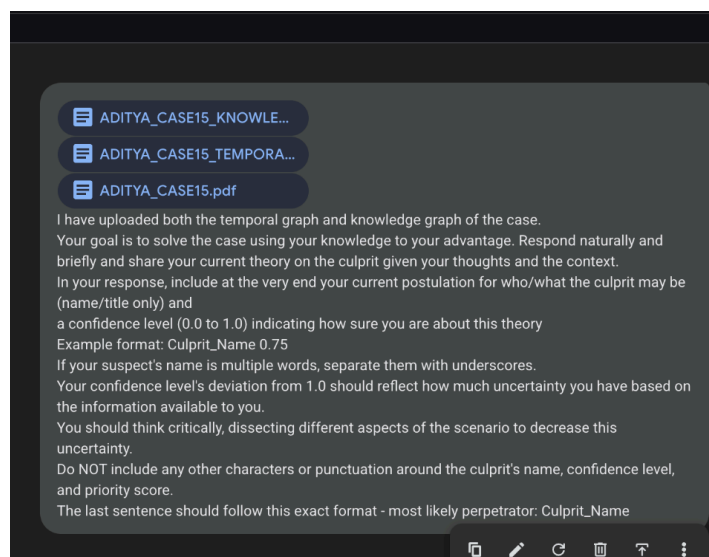


Figure 3. Sample prompt inputted to fine-tuned model

4. RESULTS

4.1 Graph-Based Reasoning Performance

Across the 24 test cases, the temporal graph condition achieved the highest accuracy of the graph-based conditions at 81.3%, slightly exceeding the raw LLM baseline of 79.2%. It did exhibit modest difficulties with overfitting, imposing time-based restrictions where they were not necessarily due. For example, in Case #2, “The Riverside Observatory Incident”, the temporal graph LLM did not recognize that the time of poison administration could differ from the victim’s time of death, causing it to preemptively rule out the correct culprit. Alternatively, the knowledge graph alone achieved 75.0%, struggling particularly with cases involving deception,

hidden intent, and indirect implication across multiple entities. The combined graph condition, despite incorporating both knowledge structures, also yielded 75.0% accuracy – performing no better than its weakest component. Both graph conditions shared a common limitation: difficulty with the global synthesis required to detect hidden intent and deception.

Differences in self-assessed confidence mirrored these accuracy differences. The temporal graph condition produced the highest average confidence at 0.908, compared to 0.894 for the raw baseline and 0.860 for the knowledge graph condition. The temporal graph's edge in both accuracy and confidence likely reflects that time-based evidence – the sequence, duration, and contradiction of events – provides more irrefutable constraints on the solution space than ontological associations alone.

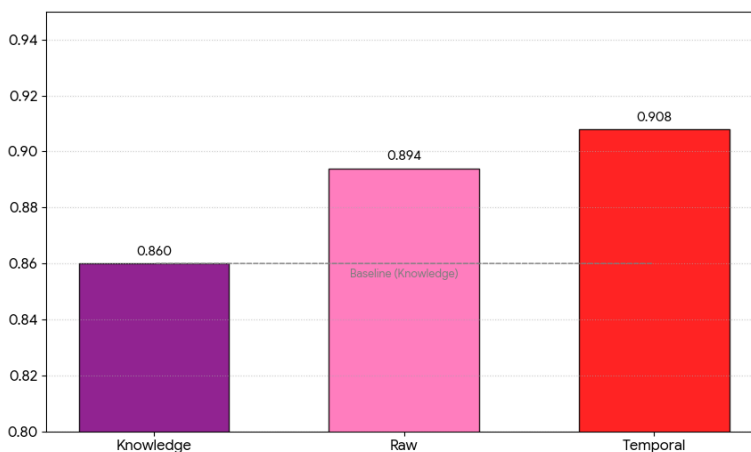


Figure 4. *Self-assessed confidence level by model*

4.2 Fine-Tuned Model Performance

The fine-tuned model demonstrated the most substantial improvements of any condition. Training accuracy reached 81.2%, with validation accuracy rising to 90.0% by the final epoch. When applied to the 24 complex test cases, the fine-tuned model achieved 89.6% accuracy – a 10.4 percentage point gain over the raw baseline – accompanied by an average self-assessed confidence of 91.0%. Validation loss was 1.29 and testing loss was 1.43, indicating reasonable generalization without severe overfitting.

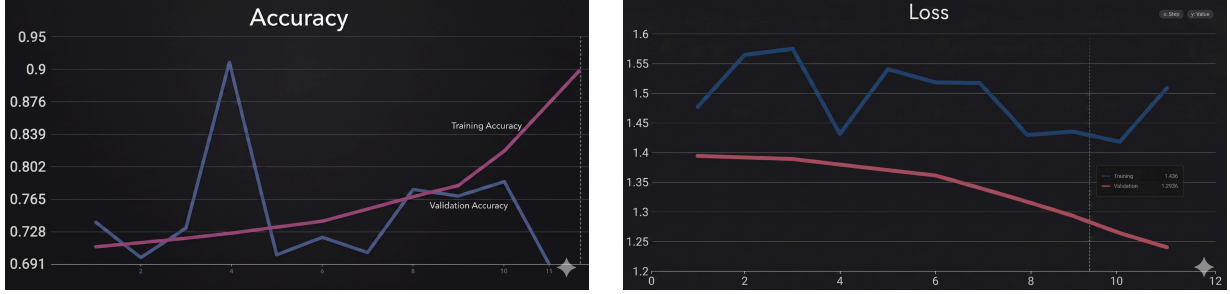


Figure 5. *Accuracy and loss functions of fine-tuned model*

Qualitatively, the fine-tuned model demonstrated remarkably more structured and systematic reasoning. In the representative test case ("The Mireille Labs Poisoning"), the fine-tuned model correctly identified Conrad Voss and James Wren as co-perpetrators by reasoning through the thallium sulfate latency period, eliminating Diane Beaumont and Dr. Ashouri on the basis of opportunity, and triangulating Voss's financial motive with Wren's material access. The model explicitly distinguished the orchestrator (Voss) from the committer (Wren) – a distinction neither the raw LLM nor the standalone graphs reliably made.

5. DISCUSSION

The central finding of this study is that an LLM's reasoning performance is sensitive not only to the volume of information it receives, but also to the structure that information is organized within. Time-based evidence, encoded as a temporal graph, proved more effective than ontological evidence for guiding multi-step deductive reasoning. This supports the hypothesis that explicitly encoding causality and sequencing – rather than merely documenting relationships – is particularly valuable for complex tasks.

The underperformance of the combined graph relative to expectations is a notable negative result. Rather than combining the strengths of both representations, the combined condition appeared to suffer the weaknesses of each. One explanation is context overload: providing both graphs simultaneously may have increased the noise in the LLM's context window, diluting clarity. Alternatively, the two graphs may have occasionally presented conflicting or redundant framings of the same evidence, confusing rather than reinforcing the LLM's reasoning.

Fine-tuning on step-by-step reasoning chains emerged as the most powerful intervention. Explicitly supervising the model to produce intermediate reasoning steps, rather than just final answers, substantially improves performance on complex reasoning tasks. The fine-tuned model's ability to integrate both graphs and produce reasoning chains grounded in explicit evidence suggests that the combination of structured inputs and targeted training is more powerful than either alone.

A recurring practical challenge was the quality and cost of LLM-generated case data. Several cases were found to contain internal inconsistencies that required revision. The temporal graph was instrumental in recognizing these contradictions. Future work using real case files, subject to appropriate ethical and legal constraints, would substantially improve the ecological validity of these findings. Additionally, expanding file material to include photographic and videographic evidence – not just descriptions of these – would help to increase applications.

6. CONCLUSION

This project developed and evaluated a pipeline for LLM-based crime-solving using two graph-based knowledge representations – a temporal graph and a knowledge graph – alongside a fine-tuned model. The temporal graph outperformed the baseline LLM, and fine-tuning on structured reasoning chains produced the most substantial accuracy gains (89.6%). These results demonstrate that information structure, not just information volume, is a key component impacting LLM reasoning performance.

Beyond criminal investigation, these findings have implications for any domain requiring multi-step deductive reasoning under uncertainty: crime, of course, but also medical diagnosis, financial fraud detection, and personal decision support. Future work could extend this framework to multimodal evidence (photographs, video), larger and more diverse case datasets, and collaborative architectures where knowledge structures are assigned to different LLMs. The question of whether LLM collaboration can outperform a single fine-tuned model on such tasks remains open and compelling.

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